

The **nycbullets** package

New York City Subway route bullets

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1 Introduction

This package typesets the route “bullets” of the New York City Subway: **A** **C** **E**, **F** **F**, **7** **7**, and so on. The artwork comes from the [New York City Subway bullets category on Wikimedia Commons](#), where nearly all of it is in the public domain (see section 3 for the exceptions). It has been compiled into a family of OpenType color fonts, one per set of bullets on Commons, plus one font containing all of them.

2 Usage

Load the package with Lua^AT_EX:

```
\usepackage{nycbullets}
```

\nycbullet	\nycbullet [<i><options></i>]{ <i><key></i> }
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Typesets the bullet named *<key>* from the current set. The *<options>* are any of the options of **\nycbulletsetup**, applied to this bullet only; as a shortcut, an option that is just the name of a set selects that set, so **\nycbullet[helvetica]{F}** gives the Helvetica-set **F**.

A *<key>* names the route, followed by modifiers and variants:

- The route itself: **F**, **7**, **SIR**, **JFK**, **10**, or **-** for the dash bullet.
- A lower-case **d** for the diamond (express) form: **Fd** **F**, **6d** **6**. A few R32 “special” forms add an **s**: **Bs**, **Bsd**.
- Variants, introduced by a dot: colors such as **M.brown** **M**, historical routings such as **B.broadway**, years such as **AA.1967-1979**, and so on.

The complete list of keys in each set is in section 4. An unknown key is an error.

\nycbulletsetup	\nycbulletsetup { <i><key=value list></i> }
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Sets options for all following bullets in the current group. The same options can be given when loading the package.

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set=*<name>* The bullet set to use. The sets are **std** (or **standard**), **helv** (**helvetica**), **nycta**, **r62a**, **r32**, **retired**, **legacy**, and **all**. The default is **std**.

scale=*<number>* Size of the bullets relative to the current font size. The default, 1, makes a circular bullet 0.8em across, a little taller than a capital letter.

raise=*<dimension>* Shift the bullets up (or down, if negative). The default is 0pt: circular bullets then extend 0.1em below the baseline.

color=**true**|**false** Whether to use the colored glyphs (the default) or the monochrome outlines.

renderer=**node**|**harfbuzz** Which luaotfload shaper to use. Both render the bullets in color.

The **color** and **renderer** options take effect when a set’s font is first used, so set them in the preamble.

\nycbulletfont \nycbulletfont[*<set>*]

Switches to the bullet font of a set, for typing keys directly. The keys are ligatures in the font, so `{\nycbulletfont F Fd SIR}` gives   . Characters that are not part of a key come out as empty boxes. Separate adjacent bullets with spaces: “AC” is not the same as “A C”.

2.1 Engines

The bullets are in color only with LuaL^AT_EX, which renders the fonts’ COLR table. With XeL^AT_EX the package works, but the bullets come out in a single color (the current text color), because XeT_EX does not support color fonts. pdfL^AT_EX cannot use the fonts at all.

2.2 Using the fonts elsewhere

The fonts NYCTSubwayBullets-*.ttf can be installed and used in any application. Besides the COLR table they have an SVG color table (for Safari, macOS and Adobe applications) and monochrome outlines (as a fallback). Each bullet can be typed as its key, provided the application applies standard ligatures, or as its Private Use Area code point, which is listed in the tables below.

3 Credits

The bullet artwork comes from Wikimedia Commons. Nearly all of it is the work of the Metropolitan Transportation Authority and is in the public domain. These bullets are used under licenses that require attribution:

- [Special \(1967-1979\)](#) by Avy9ishere, CC BY 4.0.

4 The bullets

4.1 NYC Subway Bullets Standard (**set=std**)

Bullet	Key	Code point	Description
	-	U+E000	NYCS Bullet, Standard Set - Blank Local
	-d	U+E001	NYCS Bullet, Standard Set - Blank Express
	0	U+E002	NYCS Bullet, Standard Set - 0 Local
	0d	U+E003	NYCS Bullet, Standard Set - 0 Express
	1	U+E004	NYCS Bullet, Standard Set - 1 Local
	1d	U+E005	NYCS Bullet, Standard Set - 1 Express
	2	U+E006	NYCS Bullet, Standard Set - 2 Local
	2d	U+E007	NYCS Bullet, Standard Set - 2 Express
	3	U+E008	NYCS Bullet, Standard Set - 3 Local
	3d	U+E009	NYCS Bullet, Standard Set - 3 Express
	4	U+E00A	NYCS Bullet, Standard Set - 4 Local
	4d	U+E00B	NYCS Bullet, Standard Set - 4 Express
	5	U+E00C	NYCS Bullet, Standard Set - 5 Local
	5d	U+E00D	NYCS Bullet, Standard Set - 5 Express
	6	U+E00E	NYCS Bullet, Standard Set - 6 Local
	6d	U+E00F	NYCS Bullet, Standard Set - 6 Express
	7	U+E010	NYCS Bullet, Standard Set - 7 Local
	7d	U+E011	NYCS Bullet, Standard Set - 7 Express
	8	U+E012	NYCS Bullet, Standard Set - 8 Local (R62A)
	8.turquoise	U+E013	NYCS Bullet, Standard Set - 8 Local (Third Avenue)
	8d	U+E014	NYCS Bullet, Standard Set - 8 Express (R62A)
	8d.turquoise	U+E015	NYCS Bullet, Standard Set - 8 Express (Third Avenue)
	9	U+E016	NYCS Bullet, Standard Set - 9 Local
	9d	U+E017	NYCS Bullet, Standard Set - 9 Express
	10	U+E018	NYCS Bullet, Standard Set - 10 (R62A)
	11	U+E019	NYCS Bullet, Standard Set - 11 (R62A)
	12	U+E01A	NYCS Bullet, Standard Set - 12 (R62A)
	13	U+E01B	NYCS Bullet, Standard Set - 13 (R62A)
	A	U+E01C	NYCS Bullet, Standard Set - A Local
	A.gray	U+E01D	NYCS Bullet, Standard Set - A Local
	Ad	U+E01E	NYCS Bullet, Standard Set - A Express
	Ad.gray	U+E01F	NYCS Bullet, Standard Set - A Express
	B	U+E020	NYCS Bullet, Standard Set - B Local
	Bd	U+E021	NYCS Bullet, Standard Set - B Express

Bullet	Key	Code point	Description
	Bs	U+E022	NYCS Bullet, Standard Set - B South Local (1986–1988)
	Bsd	U+E023	NYCS Bullet, Standard Set - B South Express (1986–1988)
	C	U+E024	NYCS Bullet, Standard Set - C Local
	Cd	U+E025	NYCS Bullet, Standard Set - C Express
	D	U+E026	NYCS Bullet, Standard Set - D Local
	Dd	U+E027	NYCS Bullet, Standard Set - D Express
	Ds	U+E028	NYCS Bullet, Standard Set - D South Local (1986–1988)
	Dsd	U+E029	NYCS Bullet, Standard Set - D South Express (1986–1988)
	E	U+E02A	NYCS Bullet, Standard Set - E Local
	Ed	U+E02B	NYCS Bullet, Standard Set - E Express
	F	U+E02C	NYCS Bullet, Standard Set - F Local
	Fd	U+E02D	NYCS Bullet, Standard Set - F Express
	G	U+E02E	NYCS Bullet, Standard Set - G Local
	Gd	U+E02F	NYCS Bullet, Standard Set - G Express
	H	U+E030	NYCS Bullet, Standard Set - H Local
	Hd	U+E031	NYCS Bullet, Standard Set - H Express
	I	U+E032	NYCS Bullet, Standard Set - I Local
	Id	U+E033	NYCS Bullet, Standard Set - I Express
	J	U+E034	NYCS Bullet, Standard Set - J Local
	Jd	U+E035	NYCS Bullet, Standard Set - J Express
	K	U+E036	NYCS Bullet, Standard Set - K Local
	Kd	U+E037	NYCS Bullet, Standard Set - K Express
	L	U+E038	NYCS Bullet, Standard Set - L Local
	Ld	U+E039	NYCS Bullet, Standard Set - L Express
	M	U+E03A	NYCS Bullet, Standard Set - M Local
	M.brown	U+E03B	NYCS Bullet, Standard Set - M Local (1979–2010)
	Md	U+E03C	NYCS Bullet, Standard Set - M Express
	Md.brown	U+E03D	NYCS Bullet, Standard Set - M Express (2001–2010)
	N	U+E03E	NYCS Bullet, Standard Set - N Local
	Nd	U+E03F	NYCS Bullet, Standard Set - N Express
	O	U+E040	NYCS Bullet, Standard Set - O Local
	Od	U+E041	NYCS Bullet, Standard Set - O Express
	P	U+E042	NYCS Bullet, Standard Set - P Local (R32)
	Pd	U+E043	NYCS Bullet, Standard Set - P Express (R32)
	Q	U+E044	NYCS Bullet, Standard Set - Q Local
	Q.orange	U+E045	NYCS Bullet, Standard Set - Q Local (1988–2001)

Bullet	Key	Code point	Description
	Qd	U+E046	NYCS Bullet, Standard Set - Q Express
	Qd.orange	U+E047	NYCS Bullet, Standard Set - Q Express (1988–2001)
	R	U+E048	NYCS Bullet, Standard Set - R Local
	R.brown	U+E049	NYCS Bullet, Standard Set - R Local (1985–1987)
	Rd	U+E04A	NYCS Bullet, Standard Set - R Express
	Rd.brown	U+E04B	NYCS Bullet, Standard Set - R Express (1985–1987)
	S	U+E04C	NYCS Bullet, Standard Set - Shuttles
	Sd	U+E04D	NYCS Bullet, Standard Set - Shuttle Expresses
	S6	U+E04E	NYCS Bullet, Standard Set - Sixth Avenue Shuttles (1997–2000 and 2001)
	S6d	U+E04F	NYCS Bullet, Standard Set - Sixth Avenue Shuttle Expresses (1997–2000 and 2001)
	SB	U+E050	NYCS Bullet, Standard Set - 63rd Street Shuttle (1998–1999)
	SBd	U+E051	NYCS Bullet, Standard Set - 63rd Street Shuttle Express (1998–1999)
	SF	U+E052	NYCS Bullet, Standard Set - Franklin Avenue Shuttle
	SF.silver	U+E053	NYCS Bullet, Standard Set - Franklin Avenue Shuttle (R32)
	SFd.silver	U+E054	NYCS Bullet, Standard Set - Franklin Avenue Shuttle Express (R32)
	SIR	U+E055	NYCS Bullet, Standard Set - Staten Island Railway
	SIRd	U+E056	NYCS Bullet, Standard Set - Staten Island Railway Express
	SR	U+E057	NYCS Bullet, Standard Set - Rockaway Park Shuttle
	SR.blue	U+E058	NYCS Bullet, Standard Set - Shuttles
	SRd.blue	U+E059	NYCS Bullet, Standard Set - Shuttle Expresses
	T	U+E05A	NYCS Bullet, Standard Set - T Local
	T.black	U+E05B	NYCS Bullet, Standard Set - T Local (R32)
	Td	U+E05C	NYCS Bullet, Standard Set - T Express
	Td.black	U+E05D	NYCS Bullet, Standard Set - T Express (R32)
	U	U+E05E	NYCS Bullet, Standard Set - U Local (R32)
	Ud	U+E05F	NYCS Bullet, Standard Set - U Express (R32)
	V	U+E060	NYCS Bullet, Standard Set - V Local

Bullet	Key	Code point	Description
	Vd	U+E061	NYCS Bullet, Standard Set - V Express
	W	U+E062	NYCS Bullet, Standard Set - W Local
	Wd	U+E063	NYCS Bullet, Standard Set - W Express
	X	U+E064	NYCS Bullet, Standard Set - X Local (R32)
	Xd	U+E065	NYCS Bullet, Standard Set - X Express (R32)
	Y	U+E066	NYCS Bullet, Standard Set - Y Local (R32)
	Yd	U+E067	NYCS Bullet, Standard Set - Y Express (R32)
	Z	U+E068	NYCS Bullet, Standard Set - Z Local
	Zd	U+E069	NYCS Bullet, Standard Set - Z Express

4.2 NYC Subway Bullets Helvetica (**set=helv**)

Bullet	Key	Code point	Description
	1	U+E000	NYCS-bull-trans-1
	2	U+E001	NYCS-bull-trans-2
	3	U+E002	NYCS-bull-trans-3
	4	U+E003	NYCS-bull-trans-4
	4d	U+E004	NYCS-bull-trans-4d
	5	U+E005	NYCS-bull-trans-5
	5d	U+E006	NYCS-bull-trans-5d
	6	U+E007	NYCS-bull-trans-6
	6d	U+E008	NYCS-bull-trans-6d
	7	U+E009	NYCS-bull-trans-7
	7d	U+E00A	NYCS-bull-trans-7d
	9	U+E00B	NYCS-bull-trans-9
	A	U+E00C	NYCS-bull-trans-A
	A.gray	U+E00D	NYCS-bull-trans-A gray
	A.orange	U+E00E	NYCS-bull-trans-A-Orange
	Ad	U+E00F	NYCS-bull-trans-Ad
	B	U+E010	NYCS-bull-trans-B
	B.yellow	U+E011	NYCS-bull-trans-B yellow
	Bd	U+E012	NYCS-bull-trans-Bd
	Bd.yellow	U+E013	NYCS-bull-trans-Bd yellow
	C	U+E014	NYCS-bull-trans-C
	Cd	U+E015	NYCS-bull-trans-Cd
	D	U+E016	NYCS-bull-trans-D
	D.yellow	U+E017	NYCS-bull-trans-D yellow
	Dd	U+E018	NYCS-bull-trans-Dd

Bullet	Key	Code point	Description
	E	U+E019	NYCS-bull-trans-E
	F	U+E01A	NYCS-bull-trans-F
	Fd	U+E01B	NYCS-bull-trans-Fd
	G	U+E01C	NYCS-bull-trans-G
	H	U+E01D	NYCS-bull-trans-H
	J	U+E01E	NYCS-bull-trans-J
	Jd	U+E01F	NYCS-bull-trans-Jd
	K	U+E020	NYCS-bull-trans-K
	K.gray	U+E021	NYCS-bull-trans-K-gray
	L	U+E022	NYCS-bull-trans-L
	M	U+E023	NYCS-bull-trans-M
	M.brown	U+E024	NYCS-bull-trans-M brown
	Md.brown	U+E025	NYCS-bull-trans-Md brown
	N	U+E026	NYCS-bull-trans-N
	Nd	U+E027	NYCS-bull-trans-Nd
	Q	U+E028	NYCS-bull-trans-Q
	Qd	U+E029	NYCS-bull-trans-Qd
	R	U+E02A	NYCS-bull-trans-R
	Rd	U+E02B	NYCS-bull-trans-Rd
	S	U+E02C	NYCS-bull-trans-S
	S.blue	U+E02D	NYCS-bull-trans-S blue
	S.orange	U+E02E	NYCS-bull-trans-S orange
	S.yellow	U+E02F	NYCS-bull-trans-S yellow
	SF	U+E030	NYCS-bull-trans-SF
	SIR	U+E031	NYCS-bull-trans-SIR
	SR	U+E032	NYCS-bull-trans-SR
	T	U+E033	NYCS-bull-trans-T
	V	U+E034	NYCS-bull-trans-V
	W	U+E035	NYCS-bull-trans-W
	Wd	U+E036	NYCS-bull-trans-Wd
	Z	U+E037	NYCS-bull-trans-Z
	Zd	U+E038	NYCS-bull-trans-Zd

4.3 NYC Subway Bullets NYCTA (set=nycta)

Bullet	Key	Code point	Description
	1	U+E000	NYCTA-Standard-Medium-1
	2	U+E001	NYCTA-Standard-Medium-2
	3	U+E002	NYCTA-Standard-Medium-3
	4	U+E003	NYCTA-Standard-Medium-4
	4d	U+E004	NYCTA-Standard-Medium-4d
	5	U+E005	NYCTA-Standard-Medium-5

Bullet	Key	Code point	Description
	5d	U+E006	NYCTA-Standard-Medium-5d
	6	U+E007	NYCTA-Standard-Medium-6
	6d	U+E008	NYCTA-Standard-Medium-6d
	7	U+E009	NYCTA-Standard-Medium-7
	7d	U+E00A	NYCTA-Standard-Medium-7d
	9	U+E00B	NYCTA-Standard-Medium-9
	A	U+E00C	NYCTA-Standard-Medium-A
	A.sixth	U+E00D	NYCTA-Standard-Medium-A sixth
	B	U+E00E	NYCTA-Standard-Medium-B
	B.broadway	U+E00F	NYCTA-Standard-Medium-B Broadway
	Bd	U+E010	NYCTA-Standard-Medium-Bd
	Bd.broadway	U+E011	NYCTA-Standard-Medium-Bd Broadway
	C	U+E012	NYCTA-Standard-Medium-C
	Cd	U+E013	NYCTA-Standard-Medium-Cd
	D	U+E014	NYCTA-Standard-Medium-D
	D.broadway	U+E015	NYCTA-Standard-Medium-D Broadway
	E	U+E016	NYCTA-Standard-Medium-E
	F	U+E017	NYCTA-Standard-Medium-F
	Fd	U+E018	NYCTA-Standard-Medium-Fd
	G	U+E019	NYCTA-Standard-Medium-G
	H	U+E01A	NYCTA-Standard-Medium-H
	J	U+E01B	NYCTA-Standard-Medium-J
	Jd	U+E01C	NYCTA-Standard-Medium-Jd
	K	U+E01D	NYCTA-Standard-Medium-K
	K.canarsie	U+E01E	NYCTA-Standard-Medium-K canarsie
	L	U+E01F	NYCTA-Standard-Medium-L
	M	U+E020	NYCTA-Standard-Medium-M
	M.nassau	U+E021	NYCTA-Standard-Medium-M-nassau
	Md.nassau	U+E022	NYCTA-Standard-Medium-Md nassau
	N	U+E023	NYCTA-Standard-Medium-N
	Nd	U+E024	NYCTA-Standard-Medium-Nd
	Q	U+E025	NYCTA-Standard-Medium-Q
	Q.sixth	U+E026	NYCTA-Standard-Medium-Q sixth
	Qd	U+E027	NYCTA Standard-Medium-Qd
	R	U+E028	NYCTA-Standard-Medium-R
	Rd	U+E029	NYCTA-Standard-Medium-Rd
	Rd.nassau	U+E02A	NYCTA-Standard-Medium-Rd nassau
	S	U+E02B	NYCTA-Standard-Medium-S
	S.63	U+E02C	NYCTA-Standard-Medium-S 63
	S.rockaway	U+E02D	NYCTA-Standard-Medium-S rockaway
	S.sixth	U+E02E	NYCTA-Standard-Medium-S sixth
	T	U+E02F	NYCTA-Standard-Medium-T
	V	U+E030	NYCTA-Standard-Medium-V
	W	U+E031	NYCTA-Standard-Medium-W

Bullet	Key	Code point	Description
	Wd	U+E032	NYCTA-Standard-Medium-Wd
	Z	U+E033	NYCTA-Standard-Medium-Z

4.4 NYC Subway Bullets R62A (**set=r62a**)

Bullet	Key	Code point	Description
	-	U+E000	NYCS Bullet, Standard Set - Blank Local
	1	U+E001	NYCS Bullet, Standard Set - 1 Local
	2	U+E002	NYCS Bullet, Standard Set - 2 Local
	3	U+E003	NYCS Bullet, Standard Set - 3 Local
	4	U+E004	NYCS Bullet, Standard Set - 4 Local
	5	U+E005	NYCS Bullet, Standard Set - 5 Local
	5d	U+E006	NYCS Bullet, Standard Set - 5 Express
	6	U+E007	NYCS Bullet, Standard Set - 6 Local
	6d	U+E008	NYCS Bullet, Standard Set - 6 Express
	7	U+E009	NYCS Bullet, Standard Set - 7 Local
	7d	U+E00A	NYCS Bullet, Standard Set - 7 Express
	8	U+E00B	NYCS Bullet, Standard Set - 8 Local (R62A)
	9	U+E00C	NYCS Bullet, Standard Set - 9 Local
	10	U+E00D	NYCS Bullet, Standard Set - 10 (R62A)
	11	U+E00E	NYCS Bullet, Standard Set - 11 (R62A)
	12	U+E00F	NYCS Bullet, Standard Set - 12 (R62A)
	13	U+E010	NYCS Bullet, Standard Set - 13 (R62A)
	S	U+E011	NYCS Bullet, Standard Set - Shuttles

4.5 NYC Subway Bullets R32 (**set=r32**)

Bullet	Key	Code point	Description
	A	U+E000	NYCS Bullet, Standard Set - A Local
	B	U+E001	NYCS Bullet, Standard Set - B Local
	Bd	U+E002	NYCS Bullet, Standard Set - B Express
	Bs	U+E003	NYCS Bullet, Standard Set - B South Local (1986–1988)
	Bsd	U+E004	NYCS Bullet, Standard Set - B South Express (1986–1988)
	C	U+E005	NYCS Bullet, Standard Set - C Local
	Cd	U+E006	NYCS Bullet, Standard Set - C Express
	D	U+E007	NYCS Bullet, Standard Set - D Local
	Ds	U+E008	NYCS Bullet, Standard Set - D South Local (1986–1988)

Bullet	Key	Code point	Description
	E	U+E009	NYCS Bullet, Standard Set - E Local
	F	U+E00A	NYCS Bullet, Standard Set - F Local
	G	U+E00B	NYCS Bullet, Standard Set - G Local
	H	U+E00C	NYCS Bullet, Standard Set - H Local
	J	U+E00D	NYCS Bullet, Standard Set - J Local
	Jd	U+E00E	NYCS Bullet, Standard Set - J Express
	JFK	U+E00F	NYCS-bull-trans-JFK
	K	U+E010	NYCS Bullet, Standard Set - K Local
	L	U+E011	NYCS Bullet, Standard Set - L Local
	M.brown	U+E012	NYCS Bullet, Standard Set - M Local (1979–2010)
	Md.brown	U+E013	NYCS Bullet, Standard Set - M Express (2001–2010)
	N	U+E014	NYCS Bullet, Standard Set - N Local
	P	U+E015	NYCS Bullet, Standard Set - P Local (R32)
	Q	U+E016	NYCS Bullet, Standard Set - Q Local
	Q.orange	U+E017	NYCS Bullet, Standard Set - Q Local (1988–2001)
	R	U+E018	NYCS Bullet, Standard Set - R Local
	Rd.brown	U+E019	NYCS Bullet, Standard Set - R Express (1985–1987)
	S6	U+E01A	NYCS Bullet, Standard Set - Sixth Avenue Shuttles (1997–2000 and 2001)
	SB	U+E01B	NYCS Bullet, Standard Set - 63rd Street Shuttle (1998–1999)
	SF.silver	U+E01C	NYCS Bullet, Standard Set - Franklin Avenue Shuttle (R32)
	T.black	U+E01D	NYCS Bullet, Standard Set - T Local (R32)
	U	U+E01E	NYCS Bullet, Standard Set - U Local (R32)
	V	U+E01F	NYCS Bullet, Standard Set - V Local
	Wd	U+E020	NYCS Bullet, Standard Set - W Express
	X	U+E021	NYCS Bullet, Standard Set - X Local (R32)
	Y	U+E022	NYCS Bullet, Standard Set - Y Local (R32)
	Z	U+E023	NYCS Bullet, Standard Set - Z Local

4.6 NYC Subway Bullets Retired (set=retired)

Bullet	Key	Code point	Description
	1.1967–1979	U+E000	1 (1967-1979 New York City Subway bullet)
	2.1967–1979	U+E001	2 (1967-1979 New York City Subway bullet)
	3.1967–1979	U+E002	3 (1967-1979 New York City Subway bullet)
	4.1967–1979	U+E003	4 (1967-1979 New York City Subway bullet)
	4d.helv	U+E004	NYCS-bull-trans-4d
	4d.std	U+E005	NYCS Bullet, Standard Set - 4 Express
	5.1967–1979	U+E006	5 (1967-1979 New York City Subway bullet)
	5d.helv	U+E007	NYCS-bull-trans-5d
	5d.std	U+E008	NYCS Bullet, Standard Set - 5 Express
	6.1967–1979	U+E009	6 (1967-1979 New York City Subway bullet)
	7.1967–1979	U+E00A	7 (1967-1979 New York City Subway bullet)
	8.1967–1979	U+E00B	8 (1967-1979 New York City Subway bullet)
	8.turquoise	U+E00C	NYCS Bullet, Standard Set - 8 Local (Third Avenue)
	9.helv	U+E00D	NYCS-bull-trans-9
	9.std	U+E00E	NYCS Bullet, Standard Set - 9 Local
	A.1967–1979	U+E00F	A Train (1967-1979)
	A.1967–1987	U+E010	A Train (1967-1987)
	A.orange	U+E011	NYCS-bull-trans-A-Orange
	Ad.1979–1987	U+E012	NYCS-bull-trans-Ad-(1979-1987)
	AA.1967–1979	U+E013	AA Train (1967-1979)
	AA.1979–1985	U+E014	AA Train (1979-1985)
	B.1967–1979	U+E015	B Train (1967-1979)
	B.yellow	U+E016	NYCS-bull-trans-B yellow
	Bd.helv	U+E017	NYCS-bull-trans-Bd
	Bd.std	U+E018	NYCS Bullet, Standard Set - B Express
	Bd.yellow	U+E019	NYCS-bull-trans-Bd yellow
	Bs	U+E01A	NYCS Bullet, Standard Set - B South Local (1986–1988)
	Bsd	U+E01B	NYCS Bullet, Standard Set - B South Express (1986–1988)
	C.1985–1987	U+E01C	C Train (1985-1987)
	Cd	U+E01D	NYCS Bullet, Standard Set - C Express
	Cd.pre-1987	U+E01E	C Train - diamond (pre-1987)
	CC.1967–1979	U+E01F	CC Train (1967-1979)
	CC.1979–1985	U+E020	CC Train (1979-1985)
	CCd	U+E021	CC Train - diamond

Bullet	Key	Code point	Description
	D.1967–1979	U+E022	D Train (1967-1979)
	D.yellow	U+E023	NYCS-bull-trans-D yellow
	Dd.helv	U+E024	NYCS-bull-trans-Dd
	Dd.std	U+E025	NYCS Bullet, Standard Set - D Express
	Ds	U+E026	NYCS Bullet, Standard Set - D South Local (1986–1988)
	E.1967–1979	U+E027	E Train (1967-1979)
	E.1979–1987	U+E028	E Train (1979-1987)
	EE.1967–1976	U+E029	EE Train (1967-1976)
	F.1967–1979	U+E02A	F Train (1967-1979)
	GG.1967–1979	U+E02B	GG Train (1967-1979)
	GG.1979–1985	U+E02C	GG Train (1979-1985)
	H.1985–1987	U+E02D	H Train (1985-1987)
	H.helv	U+E02E	NYCS-bull-trans-H
	H.std	U+E02F	NYCS Bullet, Standard Set - H Local
	HH.1967–1972	U+E030	HH Train (1967-1972)
	J.1973–1979	U+E031	J Train (1973-1979)
	Jd.helv	U+E032	NYCS-bull-trans-Jd
	Jd.std	U+E033	NYCS Bullet, Standard Set - J Express
	JFK	U+E034	NYCS-bull-trans-JFK
	JJ.1967–1973	U+E035	JJ Train (1967-1973)
	K.1973–1976	U+E036	K Train (1973-1976)
	K.1985–1987	U+E037	K Train (1985-1987)
	K.helv	U+E038	NYCS-bull-trans-K
	K.std	U+E039	NYCS Bullet, Standard Set - K Local
	KK.1967–1973	U+E03A	KK Train (1967-1973)
	LL.1967–1979	U+E03B	LL Train (1967-1979)
	LL.1979–1985	U+E03C	LL Train (1979-1985)
	M.1967–1979	U+E03D	M Train (1967-1979)
	M.brown.helv	U+E03E	NYCS-bull-trans-M brown
	M.brown.std	U+E03F	NYCS Bullet, Standard Set - M Local (1979–2010)
	Md.brown.helv	U+E040	NYCS-bull-trans-Md brown
	Md.brown.std	U+E041	NYCS Bullet, Standard Set - M Express (2001–2010)
	MJ.1967–1979	U+E042	MJ Train (1967-1979)
	N.1967–1979	U+E043	N Train (1967-1979)
	Nd.helv	U+E044	NYCS-bull-trans-Nd
	Nd.std	U+E045	NYCS Bullet, Standard Set - N Express
	NX.1967–1968	U+E046	NX Train (1967-1968)
	Q.orange.helv	U+E047	NYCS-bull-trans-Q orange
	Q.orange.std	U+E048	NYCS Bullet, Standard Set - Q Local (1988–2001)
	Qd	U+E049	NYCS Bullet, Standard Set - Q Express
	QB.1967–1979	U+E04A	QB Train (1967-1979)

Bullet	Key	Code point	Description
	QBd.1979-1985	U+E04B	QB Train (1979-1985)
	QJ.1967-1976	U+E04C	QJ Train (1967-1976)
	Rd	U+E04D	NYCS Bullet, Standard Set - R Express
	RJ.1967-1968	U+E04E	RJ Train (1967-1968)
	RR.1967-1979	U+E04F	RR Train (1967-1979)
	RR.1979-1985	U+E050	RR Train (1979-1985)
	RRd.yellow	U+E051	RR Train - Yellow diamond
	S.orange	U+E052	NYCS-bull-trans-S orange
	S.yellow	U+E053	NYCS-bull-trans-S yellow
	S6	U+E054	NYCS Bullet, Standard Set - Sixth Avenue Shuttles (1997-2000 and 2001)
	SB	U+E055	NYCS Bullet, Standard Set - 63rd Street Shuttle (1998-1999)
	SS.1967-1968.black	U+E056	SS Train (1967-1968, Black)
	SS.1967-1968.blue	U+E057	SS Train (1967-1968, Blue)
	SS.1967-1968.orange	U+E058	SS Train (1967-1968, Orange)
	SS.1967-1968.turquoise	U+E059	SS Train (1967-1968, Turquoise)
	SS.1967-1968.yellow	U+E05A	SS Train (1967-1968, Yellow)
	SS.1967-1979	U+E05B	SS Train (1967-1979)
	Special.1967-1979	U+E05C	Special (1967-1979)
	TT.1967	U+E05D	TT Train (1967)
	V.helv	U+E05E	NYCS-bull-trans-V
	V.std	U+E05F	NYCS Bullet, Standard Set - V Local
	Wd.helv	U+E060	NYCS-bull-trans-Wd
	Wd.std	U+E061	NYCS Bullet, Standard Set - W Express
	Zd	U+E062	NYCS-bull-trans-Zd

4.7 NYC Subway Bullets Legacy (**set=legacy**)

Bullet	Key	Code point	Description
	9.red	U+E000	NYCS-bull-trans-9-red
	Q.alt	U+E001	NYCS-bull-trans-Q-alt
	R.alt	U+E002	NYCS-bull-trans-R-alt
	SIR.2016-2017	U+E003	NYCS-bull-trans-SIR-2016-2017

4.8 NYC Subway Bullets All (**set=all**)

Bullet	Key	Code point	Description
	-	U+E000	NYCS Bullet, Standard Set - Blank Local
	-d	U+E001	NYCS Bullet, Standard Set - Blank Express

Bullet	Key	Code point	Description
	0	U+E002	NYCS Bullet, Standard Set - 0 Local
	0d	U+E003	NYCS Bullet, Standard Set - 0 Express
	1	U+E004	NYCS Bullet, Standard Set - 1 Local
	1.helv	U+E005	NYCS-bull-trans-1
	1.nycta	U+E006	NYCTA-Standard-Medium-1
	1.retired.1967-1979	U+E007	1 (1967-1979 New York City Subway bullet)
	1d	U+E008	NYCS Bullet, Standard Set - 1 Express
	2	U+E009	NYCS Bullet, Standard Set - 2 Local
	2.helv	U+E00A	NYCS-bull-trans-2
	2.nycta	U+E00B	NYCTA-Standard-Medium-2
	2.retired.1967-1979	U+E00C	2 (1967-1979 New York City Subway bullet)
	2d	U+E00D	NYCS Bullet, Standard Set - 2 Express
	3	U+E00E	NYCS Bullet, Standard Set - 3 Local
	3.helv	U+E00F	NYCS-bull-trans-3
	3.nycta	U+E010	NYCTA-Standard-Medium-3
	3.retired.1967-1979	U+E011	3 (1967-1979 New York City Subway bullet)
	3d	U+E012	NYCS Bullet, Standard Set - 3 Express
	4	U+E013	NYCS Bullet, Standard Set - 4 Local
	4.helv	U+E014	NYCS-bull-trans-4
	4.nycta	U+E015	NYCTA-Standard-Medium-4
	4.retired.1967-1979	U+E016	4 (1967-1979 New York City Subway bullet)
	4d	U+E017	NYCS Bullet, Standard Set - 4 Express
	4d.helv	U+E018	NYCS-bull-trans-4d
	4d.nycta	U+E019	NYCTA-Standard-Medium-4d
	5	U+E01A	NYCS Bullet, Standard Set - 5 Local
	5.helv	U+E01B	NYCS-bull-trans-5
	5.nycta	U+E01C	NYCTA-Standard-Medium-5
	5.retired.1967-1979	U+E01D	5 (1967-1979 New York City Subway bullet)
	5d	U+E01E	NYCS Bullet, Standard Set - 5 Express
	5d.helv	U+E01F	NYCS-bull-trans-5d
	5d.nycta	U+E020	NYCTA-Standard-Medium-5d
	6	U+E021	NYCS Bullet, Standard Set - 6 Local
	6.helv	U+E022	NYCS-bull-trans-6
	6.nycta	U+E023	NYCTA-Standard-Medium-6
	6.retired.1967-1979	U+E024	6 (1967-1979 New York City Subway bullet)
	6d	U+E025	NYCS Bullet, Standard Set - 6 Express
	6d.helv	U+E026	NYCS-bull-trans-6d
	6d.nycta	U+E027	NYCTA-Standard-Medium-6d
	7	U+E028	NYCS Bullet, Standard Set - 7 Local

Bullet	Key	Code point	Description
	7.helv	U+E029	NYCS-bull-trans-7
	7.nycta	U+E02A	NYCTA-Standard-Medium-7
	7.retired.1967-1979	U+E02B	7 (1967-1979 New York City Subway bullet)
	7d	U+E02C	NYCS Bullet, Standard Set - 7 Express
	7d.helv	U+E02D	NYCS-bull-trans-7d
	7d.nycta	U+E02E	NYCTA-Standard-Medium-7d
	8	U+E02F	NYCS Bullet, Standard Set - 8 Local (R62A)
	8.retired.1967-1979	U+E030	8 (1967-1979 New York City Subway bullet)
	8.turquoise	U+E031	NYCS Bullet, Standard Set - 8 Local (Third Avenue)
	8d	U+E032	NYCS Bullet, Standard Set - 8 Express (R62A)
	8d.turquoise	U+E033	NYCS Bullet, Standard Set - 8 Express (Third Avenue)
	9	U+E034	NYCS Bullet, Standard Set - 9 Local
	9.helv	U+E035	NYCS-bull-trans-9
	9.legacy.red	U+E036	NYCS-bull-trans-9-red
	9.nycta	U+E037	NYCTA-Standard-Medium-9
	9d	U+E038	NYCS Bullet, Standard Set - 9 Express
	10	U+E039	NYCS Bullet, Standard Set - 10 (R62A)
	11	U+E03A	NYCS Bullet, Standard Set - 11 (R62A)
	12	U+E03B	NYCS Bullet, Standard Set - 12 (R62A)
	13	U+E03C	NYCS Bullet, Standard Set - 13 (R62A)
	A	U+E03D	NYCS Bullet, Standard Set - A Local
	A.gray	U+E03E	NYCS Bullet, Standard Set - A Local
	A.helv	U+E03F	NYCS-bull-trans-A
	A.helv.gray	U+E040	NYCS-bull-trans-A gray
	A.helv.orange	U+E041	NYCS-bull-trans-A-Orange
	A.nycta	U+E042	NYCTA-Standard-Medium-A
	A.nycta.sixth	U+E043	NYCTA-Standard-Medium-A sixth
	A.retired.1967-1979	U+E044	A Train (1967-1979)
	A.retired.1967-1987	U+E045	A Train (1967-1987)
	Ad	U+E046	NYCS Bullet, Standard Set - A Express
	Ad.gray	U+E047	NYCS Bullet, Standard Set - A Express
	Ad.helv	U+E048	NYCS-bull-trans-Ad
	Ad.retired.1979-1987	U+E049	NYCS-bull-trans-Ad-(1979-1987)
	AA.retired.1967-1979	U+E04A	AA Train (1967-1979)
	AA.retired.1979-1985	U+E04B	AA Train (1979-1985)
	B	U+E04C	NYCS Bullet, Standard Set - B Local
	B.helv	U+E04D	NYCS-bull-trans-B
	B.helv.yellow	U+E04E	NYCS-bull-trans-B yellow
	B.nycta	U+E04F	NYCTA-Standard-Medium-B

Bullet	Key	Code point	Description
	B.nycta.broadway	U+E050	NYCTA-Standard-Medium-B Broadway
	B.retired.1967-1979	U+E051	B Train (1967-1979)
	Bd	U+E052	NYCS Bullet, Standard Set - B Express
	Bd.helv	U+E053	NYCS-bull-trans-Bd
	Bd.helv.yellow	U+E054	NYCS-bull-trans-Bd yellow
	Bd.nycta	U+E055	NYCTA-Standard-Medium-Bd
	Bd.nycta.broadway	U+E056	NYCTA-Standard-Medium-Bd Broadway
	Bs	U+E057	NYCS Bullet, Standard Set - B South Local (1986-1988)
	Bsd	U+E058	NYCS Bullet, Standard Set - B South Express (1986-1988)
	C	U+E059	NYCS Bullet, Standard Set - C Local
	C.helv	U+E05A	NYCS-bull-trans-C
	C.nycta	U+E05B	NYCTA-Standard-Medium-C
	C.retired.1985-1987	U+E05C	C Train (1985-1987)
	Cd	U+E05D	NYCS Bullet, Standard Set - C Express
	Cd.helv	U+E05E	NYCS-bull-trans-Cd
	Cd.nycta	U+E05F	NYCTA-Standard-Medium-Cd
	Cd.retired.pre-1987	U+E060	C Train - diamond (pre-1987)
	CC.retired.1967-1979	U+E061	CC Train (1967-1979)
	CC.retired.1979-1985	U+E062	CC Train (1979-1985)
	CCd.retired	U+E063	CC Train - diamond
	D	U+E064	NYCS Bullet, Standard Set - D Local
	D.helv	U+E065	NYCS-bull-trans-D
	D.helv.yellow	U+E066	NYCS-bull-trans-D yellow
	D.nycta	U+E067	NYCTA-Standard-Medium-D
	D.nycta.broadway	U+E068	NYCTA-Standard-Medium-D Broadway
	D.retired.1967-1979	U+E069	D Train (1967-1979)
	Dd	U+E06A	NYCS Bullet, Standard Set - D Express
	Dd.helv	U+E06B	NYCS-bull-trans-Dd
	Ds	U+E06C	NYCS Bullet, Standard Set - D South Local (1986-1988)
	Dsd	U+E06D	NYCS Bullet, Standard Set - D South Express (1986-1988)
	E	U+E06E	NYCS Bullet, Standard Set - E Local
	E.helv	U+E06F	NYCS-bull-trans-E
	E.nycta	U+E070	NYCTA-Standard-Medium-E
	E.retired.1967-1979	U+E071	E Train (1967-1979)
	E.retired.1979-1987	U+E072	E Train (1979-1987)
	Ed	U+E073	NYCS Bullet, Standard Set - E Express
	EE.retired.1967-1976	U+E074	EE Train (1967-1976)
	F	U+E075	NYCS Bullet, Standard Set - F Local
	F.helv	U+E076	NYCS-bull-trans-F
	F.nycta	U+E077	NYCTA-Standard-Medium-F

Bullet	Key	Code point	Description
	F.retired.1967-1979	U+E078	F Train (1967-1979)
	Fd	U+E079	NYCS Bullet, Standard Set - F Express
	Fd.helv	U+E07A	NYCS-bull-trans-Fd
	Fd.nycta	U+E07B	NYCTA-Standard-Medium-Fd
	G	U+E07C	NYCS Bullet, Standard Set - G Local
	G.helv	U+E07D	NYCS-bull-trans-G
	G.nycta	U+E07E	NYCTA-Standard-Medium-G
	Gd	U+E07F	NYCS Bullet, Standard Set - G Express
	GG.retired.1967-1979	U+E080	GG Train (1967-1979)
	GG.retired.1979-1985	U+E081	GG Train (1979-1985)
	H	U+E082	NYCS Bullet, Standard Set - H Local
	H.helv	U+E083	NYCS-bull-trans-H
	H.nycta	U+E084	NYCTA-Standard-Medium-H
	H.retired.1985-1987	U+E085	H Train (1985-1987)
	Hd	U+E086	NYCS Bullet, Standard Set - H Express
	HH.retired.1967-1972	U+E087	HH Train (1967-1972)
	I	U+E088	NYCS Bullet, Standard Set - I Local
	Id	U+E089	NYCS Bullet, Standard Set - I Express
	J	U+E08A	NYCS Bullet, Standard Set - J Local
	J.helv	U+E08B	NYCS-bull-trans-J
	J.nycta	U+E08C	NYCTA-Standard-Medium-J
	J.retired.1973-1979	U+E08D	J Train (1973-1979)
	Jd	U+E08E	NYCS Bullet, Standard Set - J Express
	Jd.helv	U+E08F	NYCS-bull-trans-Jd
	Jd.nycta	U+E090	NYCTA-Standard-Medium-Jd
	JFK.r32	U+E091	NYCS-bull-trans-JFK
	JJ.retired.1967-1973	U+E092	JJ Train (1967-1973)
	K	U+E093	NYCS Bullet, Standard Set - K Local
	K.helv	U+E094	NYCS-bull-trans-K
	K.helv.gray	U+E095	NYCS-bull-trans-K-gray
	K.nycta	U+E096	NYCTA-Standard-Medium-K
	K.nycta.canarsie	U+E097	NYCTA-Standard-Medium-K canarsie
	K.retired.1973-1976	U+E098	K Train (1973-1976)
	K.retired.1985-1987	U+E099	K Train (1985-1987)
	Kd	U+E09A	NYCS Bullet, Standard Set - K Express
	KK.retired.1967-1973	U+E09B	KK Train (1967-1973)
	L	U+E09C	NYCS Bullet, Standard Set - L Local
	L.helv	U+E09D	NYCS-bull-trans-L
	L.nycta	U+E09E	NYCTA-Standard-Medium-L
	Ld	U+E09F	NYCS Bullet, Standard Set - L Express
	LL.retired.1967-1979	U+E0A0	LL Train (1967-1979)
	LL.retired.1979-1985	U+E0A1	LL Train (1979-1985)
	M	U+E0A2	NYCS Bullet, Standard Set - M Local

Bullet	Key	Code point	Description
	M.brown	U+E0A3	NYCS Bullet, Standard Set - M Local (1979–2010)
	M.helv	U+E0A4	NYCS-bull-trans-M
	M.helv.brown	U+E0A5	NYCS-bull-trans-M brown
	M.nycta	U+E0A6	NYCTA-Standard-Medium-M
	M.nycta.nassau	U+E0A7	NYCTA-Standard-Medium-M-nassau
	M.retired.1967–1979	U+E0A8	M Train (1967-1979)
	Md	U+E0A9	NYCS Bullet, Standard Set - M Express
	Md.brown	U+E0AA	NYCS Bullet, Standard Set - M Express (2001–2010)
	Md.helv.brown	U+E0AB	NYCS-bull-trans-Md brown
	Md.nycta.nassau	U+E0AC	NYCTA-Standard-Medium-Md nassau
	MJ.retired.1967–1979	U+E0AD	MJ Train (1967-1979)
	N	U+E0AE	NYCS Bullet, Standard Set - N Local
	N.helv	U+E0AF	NYCS-bull-trans-N
	N.nycta	U+E0B0	NYCTA-Standard-Medium-N
	N.retired.1967–1979	U+E0B1	N Train (1967-1979)
	Nd	U+E0B2	NYCS Bullet, Standard Set - N Express
	Nd.helv	U+E0B3	NYCS-bull-trans-Nd
	Nd.nycta	U+E0B4	NYCTA-Standard-Medium-Nd
	NX.retired.1967–1968	U+E0B5	NX Train (1967-1968)
	O	U+E0B6	NYCS Bullet, Standard Set - O Local
	Od	U+E0B7	NYCS Bullet, Standard Set - O Express
	P	U+E0B8	NYCS Bullet, Standard Set - P Local (R32)
	Pd	U+E0B9	NYCS Bullet, Standard Set - P Express (R32)
	Q	U+E0BA	NYCS Bullet, Standard Set - Q Local
	Q.helv	U+E0BB	NYCS-bull-trans-Q
	Q.legacy.alt	U+E0BC	NYCS-bull-trans-Q-alt
	Q.nycta	U+E0BD	NYCTA-Standard-Medium-Q
	Q.nycta.sixth	U+E0BE	NYCTA-Standard-Medium-Q sixth
	Q.orange	U+E0BF	NYCS Bullet, Standard Set - Q Local (1988–2001)
	Q.retired.orange.helv	U+E0C0	NYCS-bull-trans-Q orange
	Qd	U+E0C1	NYCS Bullet, Standard Set - Q Express
	Qd.helv	U+E0C2	NYCS-bull-trans-Qd
	Qd.nycta	U+E0C3	NYCTA Standard-Medium-Qd
	Qd.orange	U+E0C4	NYCS Bullet, Standard Set - Q Express (1988–2001)
	QB.retired.1967–1979	U+E0C5	QB Train (1967-1979)
	QBd.retired.1979–1985	U+E0C6	QB Train (1979-1985)
	QJ.retired.1967–1976	U+E0C7	QJ Train (1967-1976)
	R	U+E0C8	NYCS Bullet, Standard Set - R Local

Bullet	Key	Code point	Description
	R.brown	U+E0C9	NYCS Bullet, Standard Set - R Local (1985–1987)
	R.helv	U+E0CA	NYCS-bull-trans-R
	R.legacy.alt	U+E0CB	NYCS-bull-trans-R-alt
	R.nycta	U+E0CC	NYCTA-Standard-Medium-R
	Rd	U+E0CD	NYCS Bullet, Standard Set - R Express
	Rd.brown	U+E0CE	NYCS Bullet, Standard Set - R Express (1985–1987)
	Rd.helv	U+E0CF	NYCS-bull-trans-Rd
	Rd.nycta	U+E0D0	NYCTA-Standard-Medium-Rd
	Rd.nycta.nassau	U+E0D1	NYCTA-Standard-Medium-Rd nassau
	RJ.retired.1967–1968	U+E0D2	RJ Train (1967–1968)
	RR.retired.1967–1979	U+E0D3	RR Train (1967–1979)
	RR.retired.1979–1985	U+E0D4	RR Train (1979–1985)
	RRd.retired.yellow	U+E0D5	RR Train - Yellow diamond
	S	U+E0D6	NYCS Bullet, Standard Set - Shuttles
	S.helv	U+E0D7	NYCS-bull-trans-S
	S.helv.blue	U+E0D8	NYCS-bull-trans-S blue
	S.helv.orange	U+E0D9	NYCS-bull-trans-S orange
	S.helv.yellow	U+E0DA	NYCS-bull-trans-S yellow
	S.nycta	U+E0DB	NYCTA-Standard-Medium-S
	S.nycta.63	U+E0DC	NYCTA-Standard-Medium-S 63
	S.nycta.rockaway	U+E0DD	NYCTA-Standard-Medium-S rockaway
	S.nycta.sixth	U+E0DE	NYCTA-Standard-Medium-S sixth
	Sd	U+E0DF	NYCS Bullet, Standard Set - Shuttle Expresses
	S6	U+E0E0	NYCS Bullet, Standard Set - Sixth Avenue Shuttles (1997–2000 and 2001)
	S6d	U+E0E1	NYCS Bullet, Standard Set - Sixth Avenue Shuttle Expresses (1997–2000 and 2001)
	SB	U+E0E2	NYCS Bullet, Standard Set - 63rd Street Shuttle (1998–1999)
	SBd	U+E0E3	NYCS Bullet, Standard Set - 63rd Street Shuttle Express (1998–1999)
	SF	U+E0E4	NYCS Bullet, Standard Set - Franklin Avenue Shuttle
	SF.helv	U+E0E5	NYCS-bull-trans-SF
	SF.silver	U+E0E6	NYCS Bullet, Standard Set - Franklin Avenue Shuttle (R32)
	SFd.silver	U+E0E7	NYCS Bullet, Standard Set - Franklin Avenue Shuttle Express (R32)
	SIR	U+E0E8	NYCS Bullet, Standard Set - Staten Island Railway
	SIR.helv	U+E0E9	NYCS-bull-trans-SIR
	SIR.legacy.2016–2017	U+E0EA	NYCS-bull-trans-SIR-2016–2017

Bullet	Key	Code point	Description
	SIRd	U+E0EB	NYCS Bullet, Standard Set - Staten Island Railway Express
	SR	U+E0EC	NYCS Bullet, Standard Set - Rockaway Park Shuttle
	SR.blue	U+E0ED	NYCS Bullet, Standard Set - Shuttles
	SR.helv	U+E0EE	NYCS-bull-trans-SR
	SRd.blue	U+E0EF	NYCS Bullet, Standard Set - Shuttle Expresses
	SS.retired.1967-1968.black	U+E0F0	SS Train (1967-1968, Black)
	SS.retired.1967-1968.blue	U+E0F1	SS Train (1967-1968, Blue)
	SS.retired.1967-1968.orange	U+E0F2	SS Train (1967-1968, Orange)
	SS.retired.1967-1968.turquoise	U+E0F3	SS Train (1967-1968, Turquoise)
	SS.retired.1967-1968.yellow	U+E0F4	SS Train (1967-1968, Yellow)
	SS.retired.1967-1979	U+E0F5	SS Train (1967-1979)
	Special.retired.1967-1979	U+E0F6	Special (1967-1979)
	T	U+E0F7	NYCS Bullet, Standard Set - T Local
	T.black	U+E0F8	NYCS Bullet, Standard Set - T Local (R32)
	T.helv	U+E0F9	NYCS-bull-trans-T
	T.nycta	U+E0FA	NYCTA-Standard-Medium-T
	Td	U+E0FB	NYCS Bullet, Standard Set - T Express
	Td.black	U+E0FC	NYCS Bullet, Standard Set - T Express (R32)
	TT.retired.1967	U+E0FD	TT Train (1967)
	U	U+E0FE	NYCS Bullet, Standard Set - U Local (R32)
	Ud	U+E0FF	NYCS Bullet, Standard Set - U Express (R32)
	V	U+E100	NYCS Bullet, Standard Set - V Local
	V.helv	U+E101	NYCS-bull-trans-V
	V.nycta	U+E102	NYCTA-Standard-Medium-V
	Vd	U+E103	NYCS Bullet, Standard Set - V Express
	W	U+E104	NYCS Bullet, Standard Set - W Local
	W.helv	U+E105	NYCS-bull-trans-W
	W.nycta	U+E106	NYCTA-Standard-Medium-W
	Wd	U+E107	NYCS Bullet, Standard Set - W Express
	Wd.helv	U+E108	NYCS-bull-trans-Wd
	Wd.nycta	U+E109	NYCTA-Standard-Medium-Wd
	X	U+E10A	NYCS Bullet, Standard Set - X Local (R32)
	Xd	U+E10B	NYCS Bullet, Standard Set - X Express (R32)
	Y	U+E10C	NYCS Bullet, Standard Set - Y Local (R32)
	Yd	U+E10D	NYCS Bullet, Standard Set - Y Express (R32)

Bullet	Key	Code point	Description
	Z	U+E10E	NYCS Bullet, Standard Set - Z Local
	Z.helv	U+E10F	NYCS-bull-trans-Z
	Z.nycta	U+E110	NYCTA-Standard-Medium-Z
	Zd	U+E111	NYCS Bullet, Standard Set - Z Express
	Zd.helv	U+E112	NYCS-bull-trans-Zd

5 nycbullets implementation

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5.1 Messages

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6 \msg_new:nnn { nycbullets } { pdftex }
7   {
8     The~nycbullets~package~needs~OpenType~fonts:~
9     use~LuaLaTeX~(or~XeLaTeX~for~monochrome~bullets).
10  }
11 \msg_new:nnn { nycbullets } { xetex }
12   { XeTeX~cannot~render~color~fonts:~the~bullets~will~be~monochrome. }
13 \msg_new:nnnn { nycbullets } { unknown-key }
14   { Unknown~bullet~key~`#1'~in~set~`#2'. }
15   { See~the~nycbullets~documentation~for~the~keys~in~each~set. }
16 \msg_new:nnn { nycbullets } { unknown-option }
17   { Unknown~option~`#1'. }
18 \msg_new:nnnn { nycbullets } { unknown-set }
19   { Unknown~bullet~set~`#1'. }
20   { The~known~sets~are:~#2. }

```

5.2 Engine check

```

21 \sys_if_engine_pdftex:T
22   {
23     \msg_critical:nn { nycbullets } { pdftex }
24   }
25 \sys_if_engine_xetex:T
26   { \msg_warning:nn { nycbullets } { xetex } }
27 \RequirePackage { fontspec }

```

5.3 Variables

```

28 \tl_new:N \l__nycbullets_set_tl
29 \fp_new:N \l__nycbullets_scale_fp
30 \dim_new:N \l__nycbullets_raise_dim
31 \bool_new:N \l__nycbullets_color_bool
32 \tl_new:N \l__nycbullets_renderer_tl
33 \tl_new:N \l__nycbullets_id_tl

```

```

34 \tl_new:N \l__nycbullets_cp_tl
35 \box_new:N \l__nycbullets_box

\g__nycbullets_sets_prop Set id → font file; any accepted set name → set id; the list of set ids (for messages).
\g__nycbullets_alias_prop
\g__nycbullets_names_seq
36 \prop_new:N \g__nycbullets_sets_prop
37 \prop_new:N \g__nycbullets_alias_prop
38 \seq_new:N \g__nycbullets_names_seq

(End of definition for \g__nycbullets_sets_prop, \g__nycbullets_alias_prop, and \g__nycbullets_names_seq.)

```

5.4 Declaring sets and keys

These are used by the generated files `nycbullets-sets.def` and `nycbullets-⟨set⟩.def`.

```

\__nycbullets_declare_set:nnn \nycbullets_declare_set:nnn {⟨id⟩} {⟨name⟩} {⟨font file⟩}
39 \cs_new_protected:Npn \nycbullets_declare_set:nnn #1#2#3
40 {
41   \prop_gput:Nnn \g__nycbullets_sets_prop {#1} {#3}
42   \prop_gput:Nnn \g__nycbullets_alias_prop {#1} {#1}
43   \prop_gput:Nen \g__nycbullets_alias_prop { \str_lowercase:n {#2} } {#1}
44   \seq_gput_right:Nn \g__nycbullets_names_seq {#1}
45   \prop_new:c { g__nycbullets_keys_#1_prop }
46   \bool_new:c { g__nycbullets_loaded_#1_bool }
47 }

```

(End of definition for `\nycbullets_declare_set:nnn`. This function is documented on page ??.)

```

\__nycbullets_declare_keys:nn \nycbullets_declare_keys:nn {⟨id⟩} { {⟨key⟩} {⟨hex code point⟩} ... }
48 \cs_new_protected:Npn \nycbullets_declare_keys:nn #1#2
49 {
50   \bool_set_false:N \l__nycbullets_pending_bool
51   \tl_map_tokens:nn {#2} { \__nycbullets_declare_key:nn {#1} }
52 }

```

`\tl_map_tokens:nn` hands over one braced item at a time, so keys and code points alternate: we remember each key until its code point arrives.

```

53 \tl_new:N \l__nycbullets_pending_key_tl
54 \bool_new:N \l__nycbullets_pending_bool
55 \cs_new_protected:Npn \__nycbullets_declare_key:nn #1#2
56 {
57   \bool_if:NTF \l__nycbullets_pending_bool
58   {
59     \prop_gput:cVn { g__nycbullets_keys_#1_prop } \l__nycbullets_pending_key_tl {#2}
60     \bool_set_false:N \l__nycbullets_pending_bool
61   }
62   {
63     \tl_set:Ne \l__nycbullets_pending_key_tl { \tl_to_str:n {#2} }
64     \bool_set_true:N \l__nycbullets_pending_bool
65   }
66 }

```

(End of definition for `\nycbullets_declare_keys:nn`. This function is documented on page ??.)

```

67 \file_input:n { nycbullets-sets.def }

```

5.5 Loading a set

`\_nycbullets_resolve_set:n` Set `\l\_nycbullets_id_tl` to the id of the set named `#1`, or raise an error and fall back to the first set.

```

68 \cs_new_protected:Npn \_nycbullets_resolve_set:n #1
69 {
70   \prop_get:NnNF \g\_nycbullets_alias_prop { \str_lowercase:n {#1} } \l\_nycbullets_id_tl
71   {
72     \msg_error:nnne { nycbullets } { unknown-set } {#1}
73     { \seq_use:Nn \g\_nycbullets_names_seq { ,~ } }
74     \tl_set:Nc \l\_nycbullets_id_tl { \seq_item:Nn \g\_nycbullets_names_seq { 1 } }
75   }
76 }

```

(End of definition for `\_nycbullets_resolve_set:n`.)

`\_nycbullets_load:V` On first use of a set, read its key table. This may happen inside a group, so everything it defines is global. We make sure `%` is a comment character, which it isn't in (for instance) `\DocInput`, and that line ends are ignored, so the file can't end the paragraph.

```

77 \cs_new_protected:Npn \_nycbullets_load:n #1
78 {
79   \bool_if:cF { g\_nycbullets_loaded_#1_bool }
80   {
81     \group_begin:
82     \char_set_catcode_comment:N \%
83     \char_set_catcode_ignore:n { 13 }
84     \file_input:n { nycbullets-#1.def }
85     \group_end:
86     \bool_gset_true:c { g\_nycbullets_loaded_#1_bool }
87   }
88 }
89 \cs_generate_variant:Nn \_nycbullets_load:n { V }

```

(End of definition for `\_nycbullets_load:V`.)

`\_nycbullets_declare_font:nn` Declare the font family `\@@_font_<id>.` With Lua_T_EX the `colr` feature turns on the color layers.

```

90 \cs_new_protected:Npn \_nycbullets_declare_font:nn #1#2
91 {
92   \clist_clear:N \l\_nycbullets_features_clist
93   \sys_if_engine luatex:T
94   {
95     \str_if_eq:VnT \l\_nycbullets_renderer_tl { harfbuzz }
96     { \clist_put_right:Nn \l\_nycbullets_features_clist { Renderer = HarfBuzz } }
97     \bool_if:NTF \l\_nycbullets_color_bool
98     { \clist_put_right:Nn \l\_nycbullets_features_clist { RawFeature = {+colr} } }
99     { \clist_put_right:Nn \l\_nycbullets_features_clist { RawFeature = {-
colr} } } }
100   }
101   \use:e
102   {
103     \exp_not:N \newfontfamily

```

```

104     \exp_not:c { __nycbullets_font_#1: }
105     { \exp_not:n {#2} }
106     [ \exp_not:V \l__nycbullets_features_clist ]
107   }
108 }
109 \clist_new:N \l__nycbullets_features_clist

```

(End of definition for __nycbullets_declare_font:nn.)

The font families are declared at the start of the document, after any `\nycbulletsetup` in the preamble. (`\newfontfamily` defines its command locally, so it cannot wait for the first bullet, which is typeset in a group.) Declaring a family does not load the font.

```

110 \cs_new_protected:Npn \__nycbullets_declare_fonts:
111 {
112   \prop_map_inline:Nn \g__nycbullets_sets_prop
113   { \__nycbullets_declare_font:nn {##1} {##2} }
114 }
115 \hook_gput_code:nnn { begindocument } { nycbullets } { \__nycbullets_declare_fonts: }

```

5.6 Options

```

116 \keys_define:nn { nycbullets }
117 {
118   set      .tl_set:N = \l__nycbullets_set_tl ,
119   set      .initial:n = std ,
120   scale    .fp_set:N = \l__nycbullets_scale_fp ,
121   scale    .initial:n = 1 ,
122   raise    .dim_set:N = \l__nycbullets_raise_dim ,
123   raise    .initial:n = 0pt ,
124   color    .bool_set:N = \l__nycbullets_color_bool ,
125   color    .initial:n = true ,
126   color    .default:n = true ,
127   renderer .choices:nn = { node , harfbuzz }
128   { \tl_set:Nc \l__nycbullets_renderer_tl { \l_keys_choice_tl } } ,
129   renderer .initial:n = node ,
130   unknown  .code:n =
131   {
132     \tl_if_blank:nTF {#1}
133     { \tl_set:Nc \l__nycbullets_set_tl { \l_keys_key_str } }
134     {
135       \msg_error:nne { nycbullets } { unknown-option }
136       { \l_keys_key_str }
137     }
138   } ,
139 }
140 \ProcessKeyOptions [ nycbullets ]

```

5.7 User commands

`\nycbullets_codepoint:nn` The (hexadecimal) code point of key #2 in set #1 (an id or alias), or nothing. Expandable; the set must have been used already.

```

141 \cs_new:Npn \nycbullets_codepoint:nn #1#2
142 {
143   \prop_item:cn

```

```

144     {
145       g__nycbullets_keys_
146       \exp_args:NNe \prop_item:Nn \g__nycbullets_alias_prop { \str_lowercase:n {#1} }
147       _prop
148     }
149     {#2}
150   }

```

(End of definition for \nycbullets_codepoint:nn. This function is documented on page ??.)

\nycbullets_bullet:n Typeset one bullet with the current options.

```

151 \cs_new_protected:Npn \nycbullets_bullet:n #1
152 {
153   \__nycbullets_resolve_set:V \l__nycbullets_set_tl
154   \__nycbullets_load:V \l__nycbullets_id_tl
155   \__nycbullets_bullet:Vn \l__nycbullets_id_tl {#1}
156 }

```

The set id is passed as an argument here so that it can be part of control sequence names: in \ExplSyntaxOn, \l_@@_id_tl _prop would be read as a single name.

```

157 \cs_new_protected:Npn \__nycbullets_bullet:nn #1#2
158 {
159   \prop_get:cnNTF { g__nycbullets_keys_#1_prop } {#2} \l__nycbullets_cp_tl
160   {
161     \mode_leave_vertical:
162     \hbox_set:Nn \l__nycbullets_box
163     {
164       \__nycbullets_select_font:n {#1}
165       \tex_char:D " \l__nycbullets_cp_tl \scan_stop:
166     }
167     \box_move_up:nn { \l__nycbullets_raise_dim } { \box_use_drop:N \l__nycbullets_box }
168   }
169   { \msg_error:nnnn { nycbullets } { unknown-key } {#2} {#1} }
170 }
171 \cs_generate_variant:Nn \__nycbullets_bullet:nn { V }
172 \cs_generate_variant:Nn \__nycbullets_resolve_set:n { V }

```

(End of definition for \nycbullets_bullet:n. This function is documented on page ??.)

__nycbullets_select_font:n Switch to the font of set #1 at the requested scale.

```

173 \cs_new_protected:Npn \__nycbullets_select_font:n #1
174 {
175   \use:c { __nycbullets_font_#1: }
176   \fontsize
177   { \fp_eval:n { \l__nycbullets_scale_fp * \use:c { f@size } } pt }
178   { \use:c { f@baselineskip } }
179   \selectfont
180 }
181 \cs_generate_variant:Nn \__nycbullets_select_font:n { V }

```

(End of definition for __nycbullets_select_font:n.)

```

\newcommand{\nycbullet}
\newcommand{\nycbulletsetup}
\newcommand{\nycbulletfont}
182 \NewDocumentCommand \nycbullet { o m }
183 {
184   \group_begin:
185     \IfValueT {#1} { \keys_set:nn { nycbullets } {#1} }
186     \nycbullets_bullet:n {#2}
187   \group_end:
188 }
189 \NewDocumentCommand \nycbulletsetup { m }
190 { \keys_set:nn { nycbullets } {#1} }
191 \NewDocumentCommand \nycbulletfont { o }
192 {
193   \IfValueT {#1} { \tl_set:Nn \l__nycbullets_set_tl {#1} }
194   \__nycbullets_resolve_set:V \l__nycbullets_set_tl
195   \__nycbullets_load:V \l__nycbullets_id_tl
196   \__nycbullets_select_font:V \l__nycbullets_id_tl
197 }

```

(End of definition for `\nycbullet`, `\nycbulletsetup`, and `\nycbulletfont`. These functions are documented on page 1.)

198 `\endpackage`

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